

GURBAJ SINGH

EXPERIENCE

Sabudh Foundation - Mohali

Intern

Jul 2025 – Dec 2025

Data Science

- Leveraged Python and data analytics libraries (Pandas, NumPy) to process and analyze large datasets, deriving actionable insights for predictive modelling projects.
- Performed data cleaning, feature engineering, exploratory data analysis, and model building for analytical projects using Dataiku.
- Developed the “Integrated Learning Recommendation Engine + Personalized PYLP Report Card” using SQL database, Python and Streamlit to generate personalized learning paths and downloadable performance reports.
- Collaborated in an agile environment and documented workflows for reproducibility and team knowledge sharing.

Bizzrise TECH INC – Ludhiana, Punjab

Jun 2025 – Jul 2025

Web Development

- Assisted in designing and developing responsive web pages using HTML, CSS, JavaScript, and modern UI frameworks.
- Worked on improving website performance, optimizing layouts, and enhancing user experience.
- Collaborated with senior developers to fix bugs, integrate APIs, and ensure seamless front-end functionality.
- Gained hands-on experience with version control (Git), debugging tools, and cross-browser compatibility.

PROJECTS

AI-Powered Recommendation Engine + Personalized & Downloadable PYLP Report Card

Lead Developer

2025

- Engineered a LLM based python web application integrating **Artificial Intelligence** (LLM) to generate automatic insights and report cards based on user database (MySQL).
- Coordinated MySQL data flow with the frontend interface, ensuring clean, structured user data reaches the LLM for accurate and real-time recommendations
- Integrated an intelligent chatbot interface to streamline user interactions and simplify data analysis workflows.
- **Impact:** Delivered an automated, AI-driven recommendation pipeline that cut manual analysis time by over 70%, improved data-driven recommendations, and enabled real-time, scalable insights generation for all users.

Colon Acuity AI

Lead Developer

2025

- Used Deep learning and explainable AI.
- Developed CNN-based model for colon cancer classification using histopathological images.
- Integrated XAI techniques with attention maps for enhanced diagnostic interpretability.

VisionTranslate — AI-Based Real-Time Translation Browser Extension

- Built a browser extension for real-time screen text and audio translation using OCR, speech recognition, and machine translation.
- Implemented overlay-based translated subtitles, hover translation, and text-to-speech accessibility features.
- Integrated multilingual audio transcription using Vosk and SenseVoice with real-time subtitle rendering.

Project: WhatsApp Automation & Bulk Messaging System

Lead Developer

2026

- Developed an automated WhatsApp chatbot and bulk messaging platform using Python and the [Meta WhatsApp Cloud API](#).
- Deployed backend services on [Google Cloud Run](#) with [PostgreSQL](#) database integration.
- Integrated [Google Sheets API](#) for real-time data synchronization and lead management.
- Implemented REST APIs, automated workflows, and customer communication features.

Advanced Hybrid Rag System

Lead Developer

2026

[Live Deployment](#) | [GitHub](#)

- Architected a Hybrid Retrieval-Augmented Generation (RAG) system processing 145,000+ unstructured document chunks.
- Implemented a multi-stage retrieval pipeline combining semantic vector search (Qdrant), lexical keyword matching (Apache Solr), and structural graph expansion (Neo4j).
- Integrated Reciprocal Rank Fusion (RRF) and Cohere API for neural cross-encoder reranking to optimize context relevance.
- Utilized NVIDIA Nemotron-3-Ultra-550B LLM to generate highly accurate, hallucination-resistant answers with precise citations dynamically hot-loaded during runtime.
- Containerized the application stack using Docker and deployed it on Oracle Cloud Infrastructure (OCI).
- Achieved a faithfulness score of 0.85 by validating against a 100-question domain-specific Golden Dataset.

SKILLS

Technical Skills

Python (Data Science), C & C++ Language, MySQL, HTML5, CSS3, JavaScript (Basics), Data Analytics (Dataiku), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Vector Databases (Qdrant), Graph Databases (Neo4j), Lexical Search (Apache Solr), Video/Photo Editing

Tools & Platforms

Git, GitHub, VS Code, Google Colab, Dataiku, Firebase, Gemini CLI, Docker, Oracle Cloud Infrastructure (OCI).

Management Skills

Time Management, Communication Skills, Project Management, Team-Cooperation

Soft Skills

Problem Solving, Analytical Thinking, Project Management, Adaptability, Patience

Languages

Fluent in English; Punjabi and Hindi

CERTIFICATIONS

- **Data Science Internship Certificate** – Sabudh Foundation
- **Dataiku Certifications**(Core Designer, Advanced Designer, ML Practitioner, MLOps Practitioner, Generative AI Practitioner, Developer) – Dataiku
- **Python using AI Certificate** – AI for Techies
- **Data Analytics Job Simulation** – Deloitte Australia
- **Web Development Certificate** – Bizzrise TECH INC

EDUCATION

GURU NANAK DEV ENGINEERING COLLEGE – Ludhiana, Punjab Bachelor of Technology in Computer Science Engineering	2022 - 2026
AKAL ACADEMY JAND SAHIB Higher Secondary (Class XII), CBSE	2021-2022
AKAL ACADEMY JAND SAHIB Secondary Schooling (Class X), CBSE	2019-2020

PERSONAL INFORMATION

Ludhiana, India | www.linkedin.com/in/gurbaj-singh-0309ag | github.com/gurbaajsingh9

ACHIEVEMENTS

Member of SCIE (2022-2023)